

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2017

Subject: 392153 Chemistry III

Section: 15-16

Date: 04 October 2017

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 9 pages (including this page), 9 questions and a total score of 65 points.
 2. Write all your solutions and answers on this examination sheet.
 3. This is a closed book examination.
 4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
 5. You are not allowed to open the exam paper to start to answer before the proctor's permission.
 6. You are not allowed to use the restroom during the exam except in case of an emergency.
 7. No documents are allowed to be taken out of the examination room.
 8. Calculators are not allowed in the examination.
 9. Electronic communication devices are NOT allowed in the examination room.
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**Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the University.**

INSTRUCTIONS: Written Response

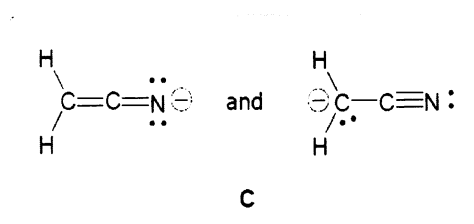
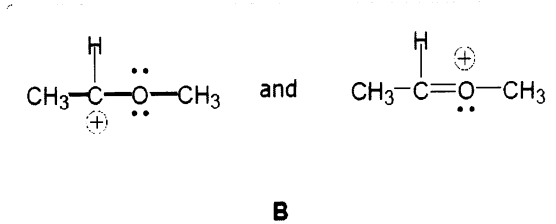
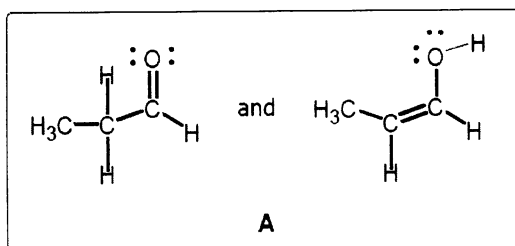
Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. **For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.**

1. When phosphorus reacts with excess chlorine gas, the compound phosphorus pentachloride (PCl_5) is formed. In the gaseous and liquid states, this substance consists of PCl_5 molecules, but in the solid state it consists of a 1:1 mixture of PCl_4^+ and PCl_6^- ions. **Predict the geometric structures of PCl_5 , PCl_4^+ and PCl_6^- .** **Suggestion:** Using Lewis structure and giving a rational explanation to support your prediction and giving a rational explanation to support your prediction. (10 marks)

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3. Which of the following choices represent(s) a pair of resonance forms?

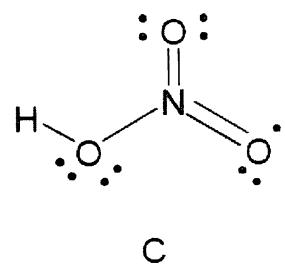
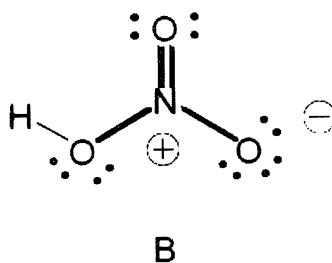
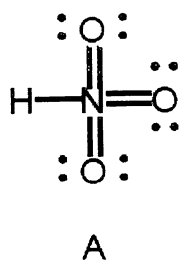
Why? (5 marks)



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4. Which of the following are correct Lewis structures for nitric acid, HNO_3 ?

Why? (5 marks)



5. Which molecule has a net dipole moment? Please give a scientific reason.

Given: H_2S NHF_2 BF_3 (5 marks)

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6. Classify the following statements as true (T) or false (F). For those statements that are false, alter them to be correct. (5 marks)

6.1 A molecule must contain two different elements joined together.

6.2 Elements tend to undergo chemical reactions to form compounds in order for their atoms to obtain noble gas configuration.

6.3 The noble gases exist as diatomic molecules.

6.4 Elements in the same group will always form the same number of covalent bonds.

7. Motor oil largely consists of molecules that consist of long chains of carbon atoms with hydrogen atoms attached to them. Using your knowledge of intermolecular forces, why wouldn't it be better to use a compound like glycerol. The formula of glycerol is $\text{CHOH}(\text{CH}_2\text{OH})_2$. (5 marks)

8. Is the reaction of methane with chlorine and fluorine to give Freon-12 (CF_2Cl_2) exothermic or endothermic reaction. Why? (10 marks)

Note: using the bond energies listed in the Table 1.



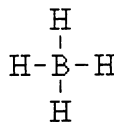
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Table 1: Average bond energies (kJ/mol)

Single Bonds				Multiple Bonds			
H—H	432	N—H	391	I—I	149	C = C	614
H—F	565	N—N	160	I—Cl	208	C ≡ C	839
H—Cl	427	N—F	272	I—Br	175	O = O	495
H—Br	363	N—Cl	200			C = O*	745
H—I	295	N—Br	243	S—H	347	C ≡ O	1072
		N—O	201	S—F	327	N = O	607
C—H	413	O—H	467	S—Cl	253	N = N	418
C—C	347	O—O	146	S—Br	218	N ≡ N	941
C—N	305	O—F	190	S—S	266	C ≡ N	891
C—O	358	O—Cl	203			C = N	615
C—F	485	O—I	234	Si—Si	340		
C—Cl	339			Si—H	393		
C—Br	276	F—F	154	Si—C	360		
C—I	240	F—Cl	253	Si—O	452		
C—S	259	F—Br	237				
		Cl—Cl	239				
		Cl—Br	218				
		Br—Br	193				

*C = O(CO₂) = 799

the structure shown below, identify



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PERIODIC TABLE OF ELEMENTS

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18

Atomic #		Symbol		Name		Weight	
1	H	Hydrogen	1.008				
3	Li	Lithium	6.94				
4	Be	Beryllium	9.0122				
11	Na	Sodium	22.990				
12	Mg	Magnesium	24.305				
19	K	Potassium	39.098				
20	Ca	Calcium	40.078				
37	Rb	Rubidium	85.468				
38	Sr	Strontium	87.62				
55	Cs	Caesium	132.91				
87	Fr	Francium	(223)				
88	Ra	Radium	(226)				
C Solid							
Hg Liquid							
H Gas							
Rf Unknown							
Metals							
Lanthanoids (Lanthanides)							
Actinoids (Actinides)							
Alkaline earth metals							
Alkali metals							
Transition metals							
Post-transition metals							
Nonmetals							
Other nonmetals							
Noble gases							
Metalloids							

Nonmetals

Other nonmetals

Noble gases

Metalloids

Metals

Alkali metals

Alkaline earth metals

Transition metals

Post-transition metals

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

57	La	Lanthanum	138.91
58	Ce	Cerium	140.12
59	Pr	Praseodymium	140.91
60	Nd	Neodymium	144.24
61	Pm	Promethium (145)	
62	Sm	Samarium	150.36
63	Eu	Europium	151.96
64	Gd	Gadolinium	157.25
65	Tb	Terbium	158.93
66	Dy	Dysprosium	162.50
67	Ho	Holmium	164.93
68	Er	Erbium	167.26
69	Tm	Thulium	168.93
70	Yb	Ytterbium	173.05
71	Lu	Lutetium	174.97
89	Ac	Actinium	(227)
90	Th	Thorium	232.04
91	Pa	Protactinium	231.04
92	U	Uranium	238.03
93	Np	Neptunium	(237)
94	Pu	Plutonium	(244)
95	Am	Americium	(243)
96	Cm	Curium	(247)
97	Bk	Berkelium	(247)
98	Cf	Californium	(251)
99	Es	Einsteinium	(252)
100	Fm	Fermium	(257)
101	Md	Mendelevium	(258)
102	No	Nobelium	(259)
103	Lr	Lawrencium	(260)

